ALMA MATER STUDIORUM UNIVERSITA' DI BOLOGNA
Second Cycle Degree in DIGITAL HUMANITIES AND DIGITAL KNOWLEDGE

# An Action-Centric Approach to Structuring Historical Events: The ACT Ontology, Taxonomy and Knowledge Graph

## FINAL THESIS PRESENTATION

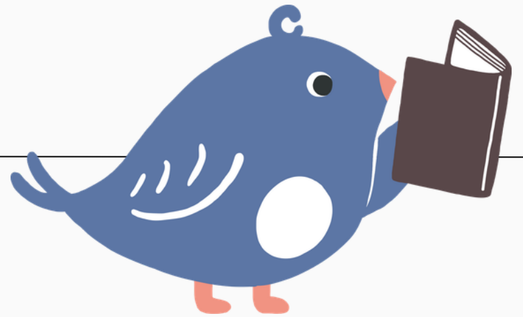
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30th October, 2026

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Co supervisor: Prof. Marilena Daquino

AGENDA


| | |
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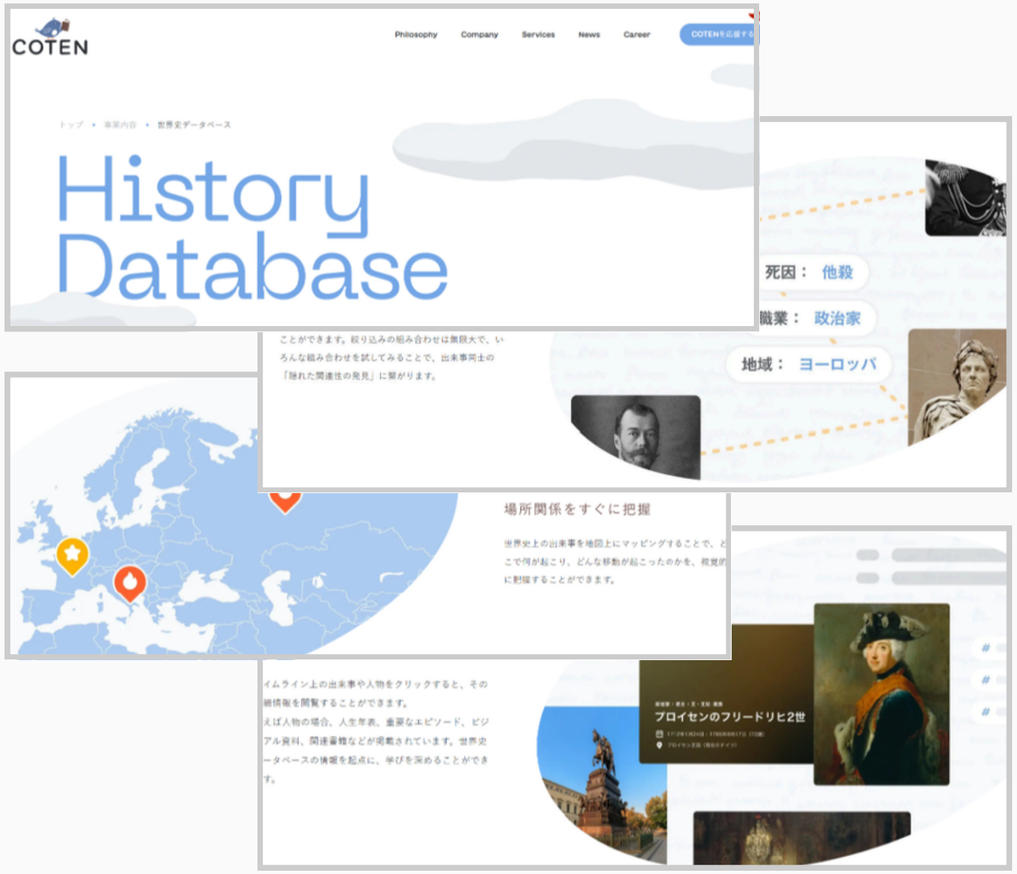



We often hear the saying, “learn from history”

. . . What do we learn from it, and how do we do that?



# The History Database Project at COTEN




- A Japanese startup company (where I work as an intern)
- Mission:  Bridging humanities knowledge and society
- Core project: the History Database Project (COTEN DB)

The main activities:

**structuring resources of history into a database.**

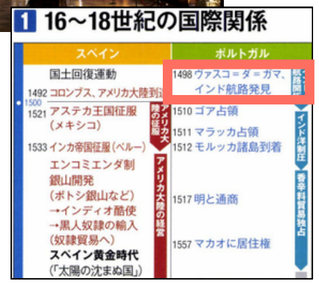
This thesis builds on the works and data of COTEN

# Historical event data extracted as entities and decomposed into “roles”

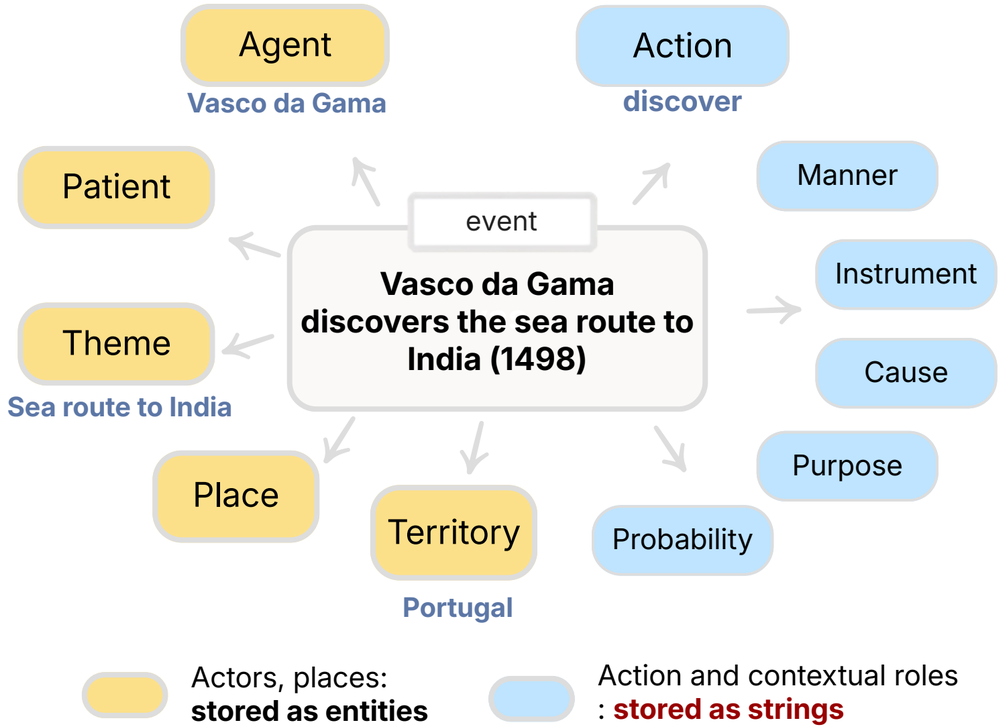
## Preliminary work at COTEN




internal OCR tool

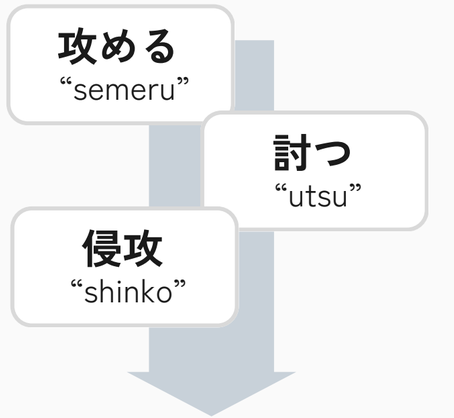



Source: chronology from history textbook




## Current situation

**Diverse action expressions are distinct. thus not searchable..**




**expressions that share menaning can be normalised**

**The “action” role is stored as a literal value, as a predicate of an event entity.**

**To make them searchable and analysable, actions also need a structured representation.**

# The Research Problem and three pillars of approach



**Research Problem**

Provide a novel perspective for analysing, classifying and studying historical events, through focusing on the **action** of individual **events** and **normalising them into structured data**.

| | Research Questions | Approach | |
|---|---|---|---|
| **RQ1** | How can a vocabulary of actions be normalised into a taxonomy, and formally represented within an ontology? | Developing the ontology, taxonomy, and initial KG | fundamental |
| **RQ2** | How can the taxonomy be validated without external reference data and with a single annotator? | Benchmarking & refining the taxonomy | extension |
| **RQ3** | How can the taxonomy run in a maintainable pipeline, and what analyses can be done from the graphs? | Building the pipeline, graph population and analysis | |

# Ontology designed with an additional layer of “action types”

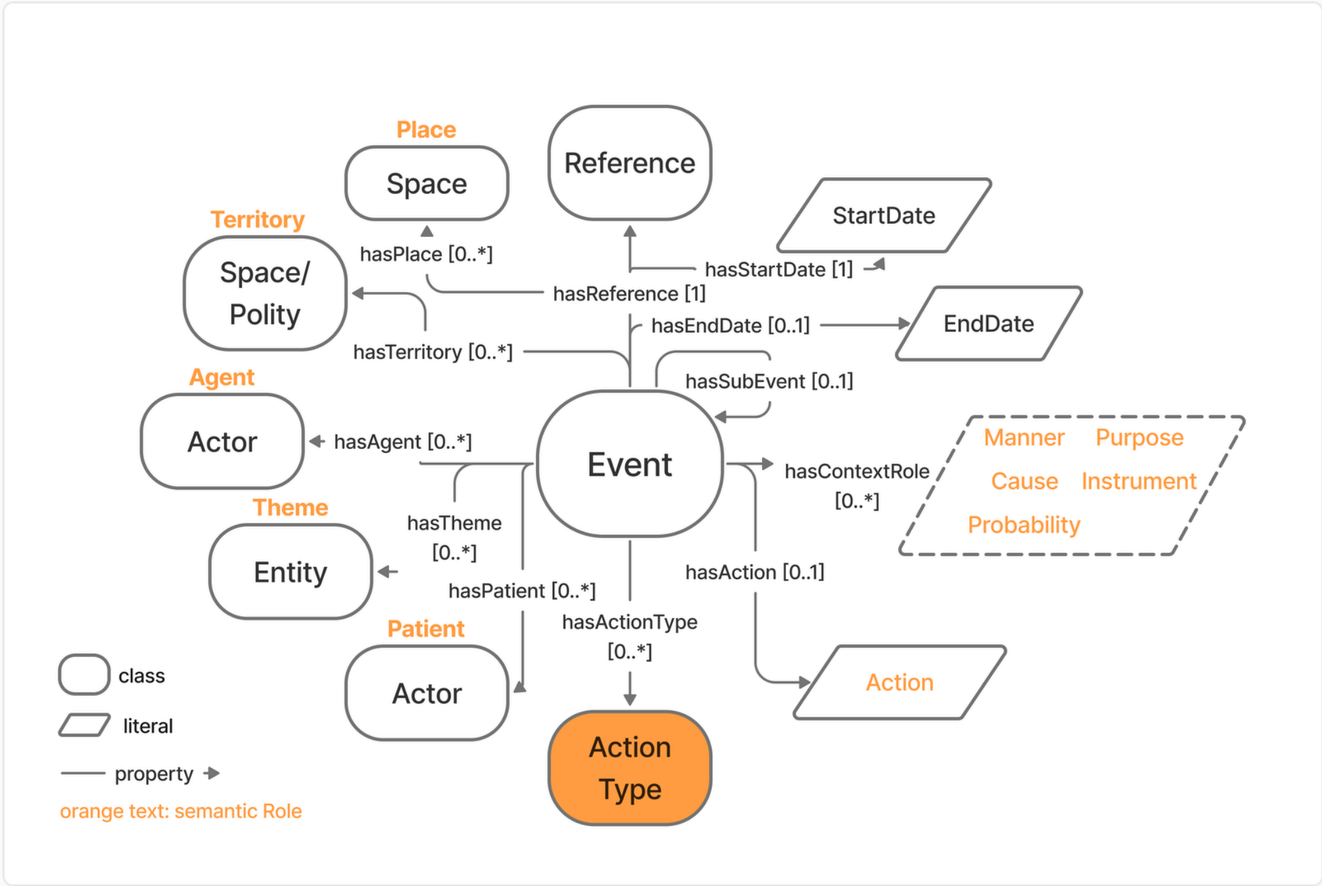


- Grounded in COTEN’s data model
- ActionType defined as a class
- Aligned using SKOS with:

**Simple Event Model / CIDOC-CRM/ Wikidata Properties**
Van Hage, W. R et al (2011)

Some examples of classes & properties mapping

| Kind | Term | External term | SKOS |
|---|---|---|---|
| Class | :ActionType | crm:E55_Type | broad |
| Class | :Actor | sem:Actor, crm:E39_Actor | close |
| Class | :Event | sem:Event, crm:E5_Event | close |
| Property | :hasActionType | crm:P2_hasType | broad |
| Property | :hasPlace | sem:hasPlace,crm:P7_took_place_at | close |




full ontology available at: <https://w3id.org/act/ontology>

# Taxonomy of actions built bottom-up from 900 expressions



**~900**

**action expressions from 7,000+ event records**

1. Pictoral of World History (詳説世界史図録)
2. Encyclopedia of C20 conflicts (20世紀紛争事典)

›

**BERT + K-means**

**semantic clustering**

Vectorizing and embedding with Tohoku Japanese BERT Clustering with Kmeans (initial: k=30)

›

**Manual + LLM coordination**

**iterative refinement**

Refining taxonomy manually with frequent example terms, mapping dictionary with LLMs

›

**37 / 8**

**ActionTypes / categories**

Military, diplomatic, political, religious ...

**Human in the loop:** **Adjusting and decisioning of the taxonomy with representative examples from the corpus, creating a dictionary that maps action expressions and action types**

# First version of taxonomy: 8 upper categories and 37 action types



1

### Military Acts & Suppression

MilitaryAttack · Conflict· Conquest · Victory · Defeat · Massacre · Suppression

2

### Resistance & Revolt

Resistance · Insurrection

3

### Diplomacy & Relations

Negotiation · Agreement · Declaration · Order · Rupture

4

### Political & Institutional

Accession · Abdication · Legislation

5

### Formations & Endings

Foundation · Beginning · Ending · Dissolution · Death · Independence · Release

6

### Movement & Transformation

Movement · Transformation · Escalation · Revival

7

### Culture, Production & Knowledge

Discovery · Construction · Invention · CulturalExpression · TradeEconomy

8

### General & Miscellaneous

Success · Failure · General

full & revised taxonomy available at: <https://w3id.org/act/taxonomy>

# Benchmarking with different classifiers to assess current state



**Ground truth**
The author's assignments for 492 (478) action expressions in 37 (33) action types*

**Classifiers**
From a majority baseline to four LLMs acting as annotators, given only the type names and representative expressions

**Evaluation protocol**
Stratified five-fold cross-validation for classifiers using the ground truth

**Metric**
Macro-F1 as primary + accuracy
Fleiss κ for agreements among LLMs

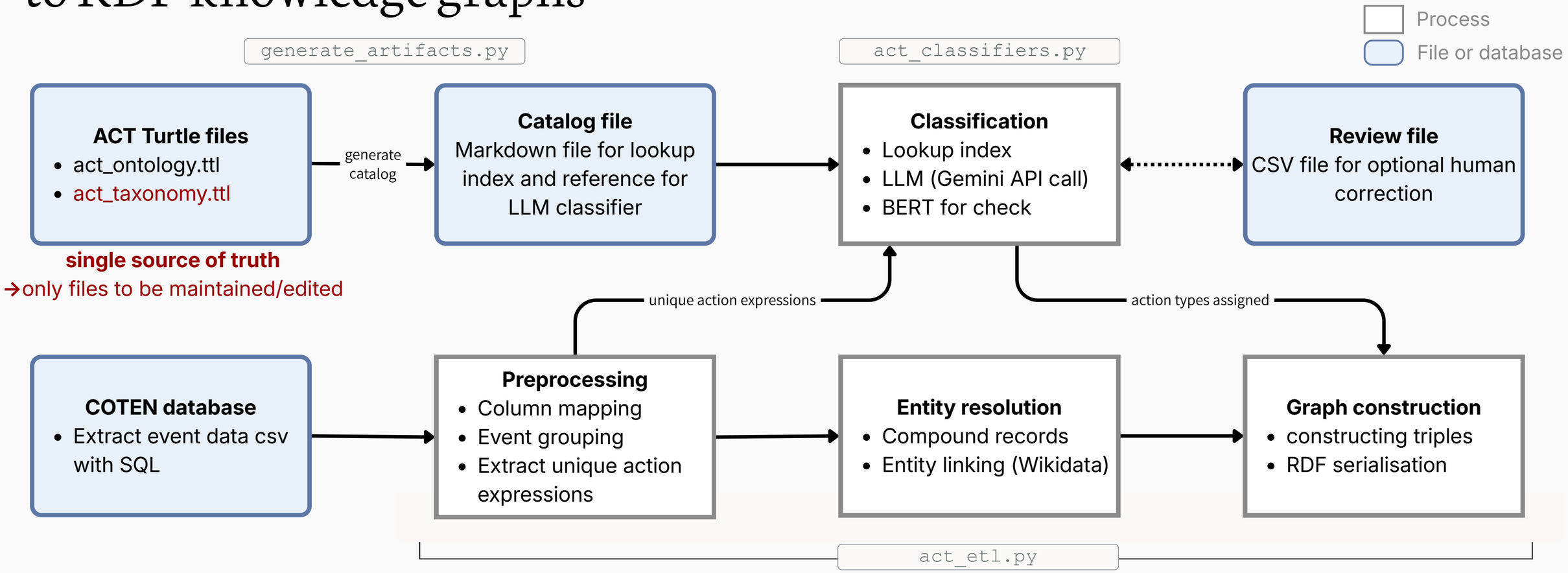
| Classifier | How it works | Macro-F1 |
|---|---|---|
| Majority Class | Always assigns the most frequent type | 0.6% |
| Character n-gram | TF-IDF over character sequences | 19.9% |
| Japanese WordNet | Path similarity to the representative terms of each type | 47.0% |
| Tohoku (Japanese) BERT | Nearest centroid of the training-fold embeddings | 51.5% |
| Large Language Models | Zero-shot prompt of type names and representative expressions | Gemini fl. 3.5 - **82.1%**<br>Fable 5 - 82.0%<br>Sonnet 4.6 - 80.3%<br>ChatGPT-4o - 78.4% |

Fleiss κ among the four LLM models : **0.84**

**The disagreements / unanimous wrong results from LLMs led to the revision of the taxonomy from 37 to 44 types.**

*numbers in () are the action expressions and types after excluding types with fewer than five expressions, which could not be included in the evaluation folds

# A pipeline that extracts event data, assigns action types and converts to RDF knowledge graphs

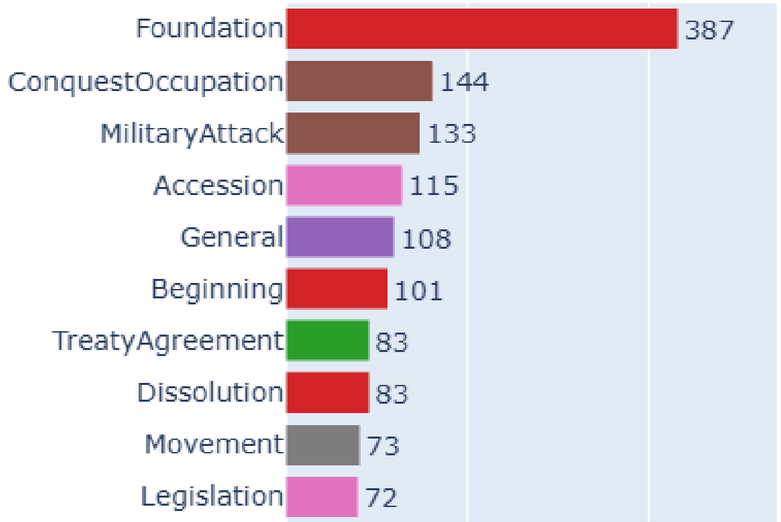


# “Foundation” is the most frequent action type for all three sources

Profile of each source and their 10 most frequent action types

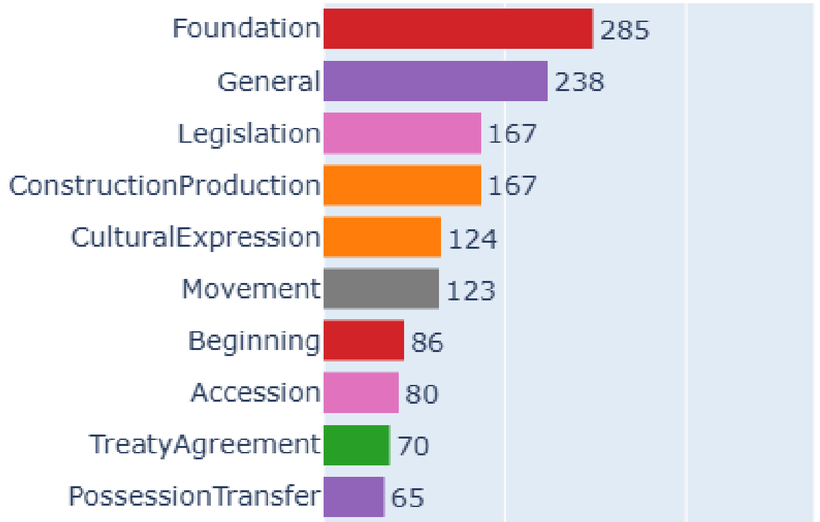
1. ***Pictorial of World History***

**39603** triples
**3390** event records
**1788** events with an action



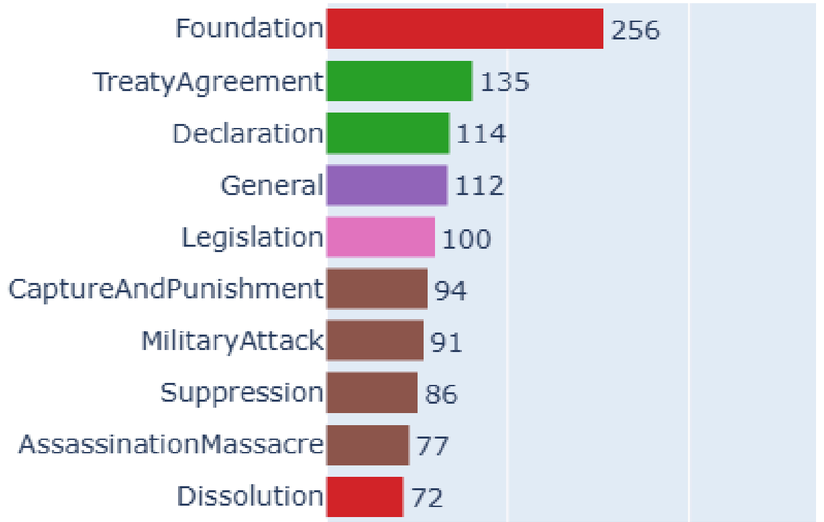

2. ***Pictorial of Japanese History***

**38984** triples
**3222** event records
**1912** events with an action




3. ***Encyclopedia of 20C World Conflicts***

**42086** triples
**3671** event records
**1671** events with an action







**The knowledge graph can answer questions starting from an action type,**
**combined with other event related features such as temporal, locations, people, polities etc.**

# ACT provides a perspective for analysing, classifying and studying historical events through actions

## Contributions



**Adds a layer of semantic depth previous works have not provided.**

**Opens path to action-oriented, large-scale analysis of historical events.**

## Outcomes

**RQ1.** A reusable taxonomy of action types and an OWL ontology aligned with existing models

**RQ2.** A means to evaluate a single-annotated taxonomy, revised taxonomy guided by LLM annotators

**RQ3.** A maintainable pipeline with a classifier and conversion to RDF / KG of three sources

## Limitations and Future work

- Limitations include dependence on preliminary works of COTEN for sources, bias in Japanese language.
- Future works include expanding the graph, and further analysis with more data.



We often hear the saying, “learn from history”

. . . What do we learn from it, and how do we do that?

A shared vocabulary of actions may help us approach this question



# Thank you!

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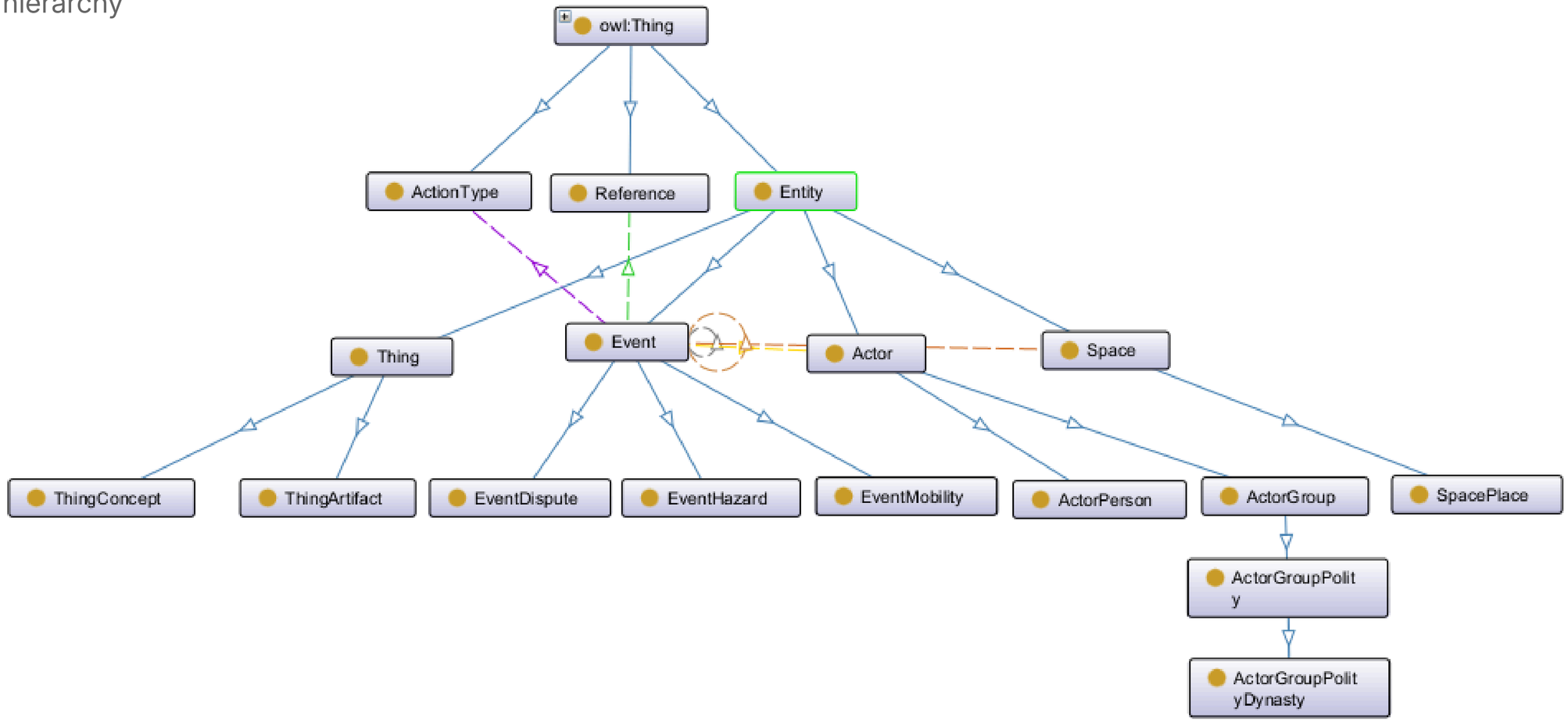


ACT GitHub repo

# APPENDIX

# RQ1 - the ONTOLOGY

Class hierarchy



# Revised version of taxonomy: 8 categories and 44 action types

1

### Military, Violence & Suppression

MilitaryAttack · Conflict· Conquest · Victory · Defeat · Massacre · Suppression · Subjugation · Capture

2

### Resistance & Revolt

Resistance · Insurrection

3

### Diplomacy & Relations

Negotiation · Agreement · Declaration · Order · Rupture · Authorisation · Communication ·Rejection

4

### Political & Institutional

Accession · Abdication · Legislation

5

### Existence & Lifecycle

Foundation · Beginning ·Dissolution · Death · Independece

6

### State & Transition

Movement · Transformation · Escalation · Revival · Deescalation · Release · Suspension · Repetition

7

### Culture, Production & Knowledge

Discovery · Construction · Invention · CulturalExpression · TradeEconomy

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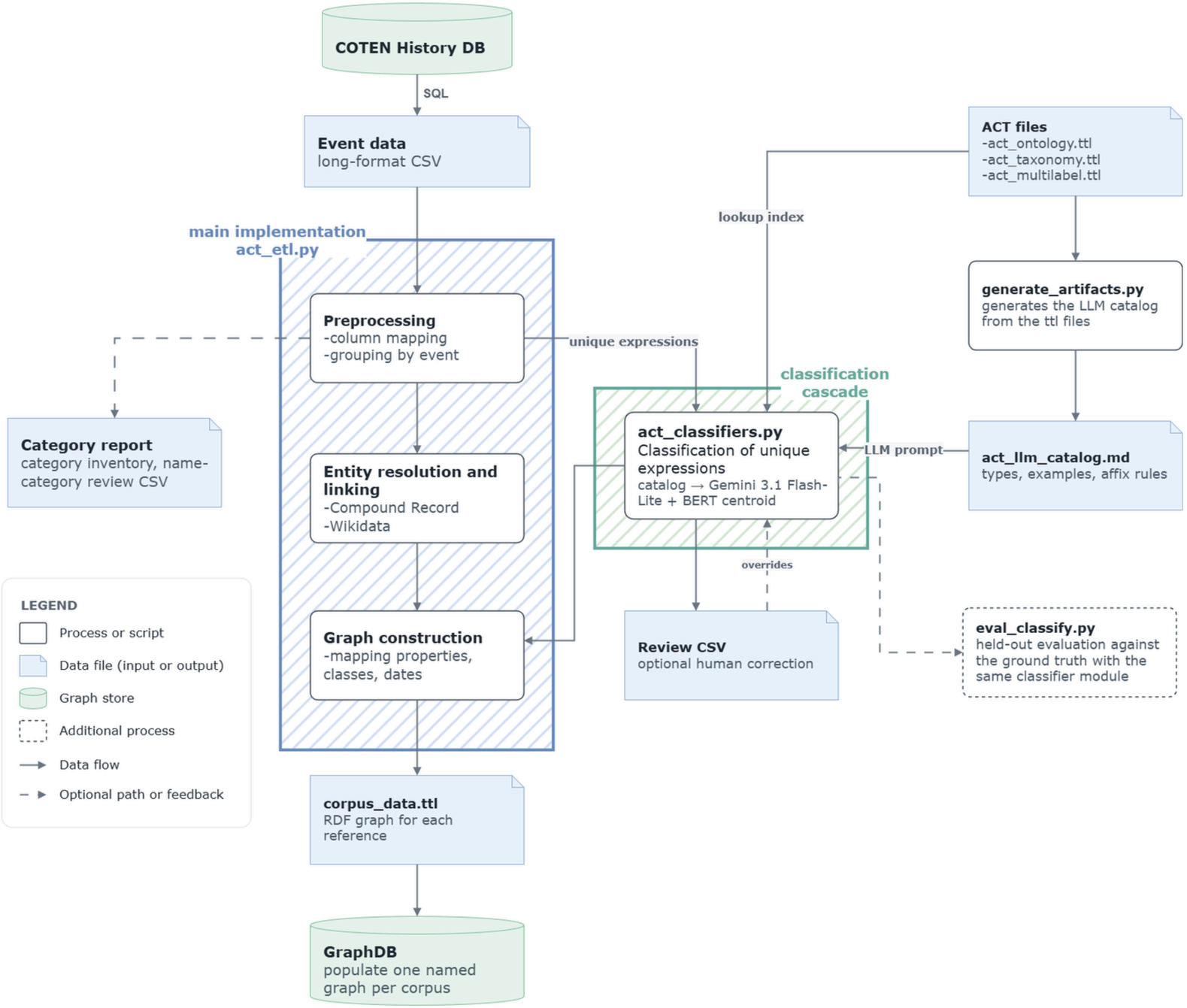
### General & Miscellaneous

Success · Failure · General · Possession

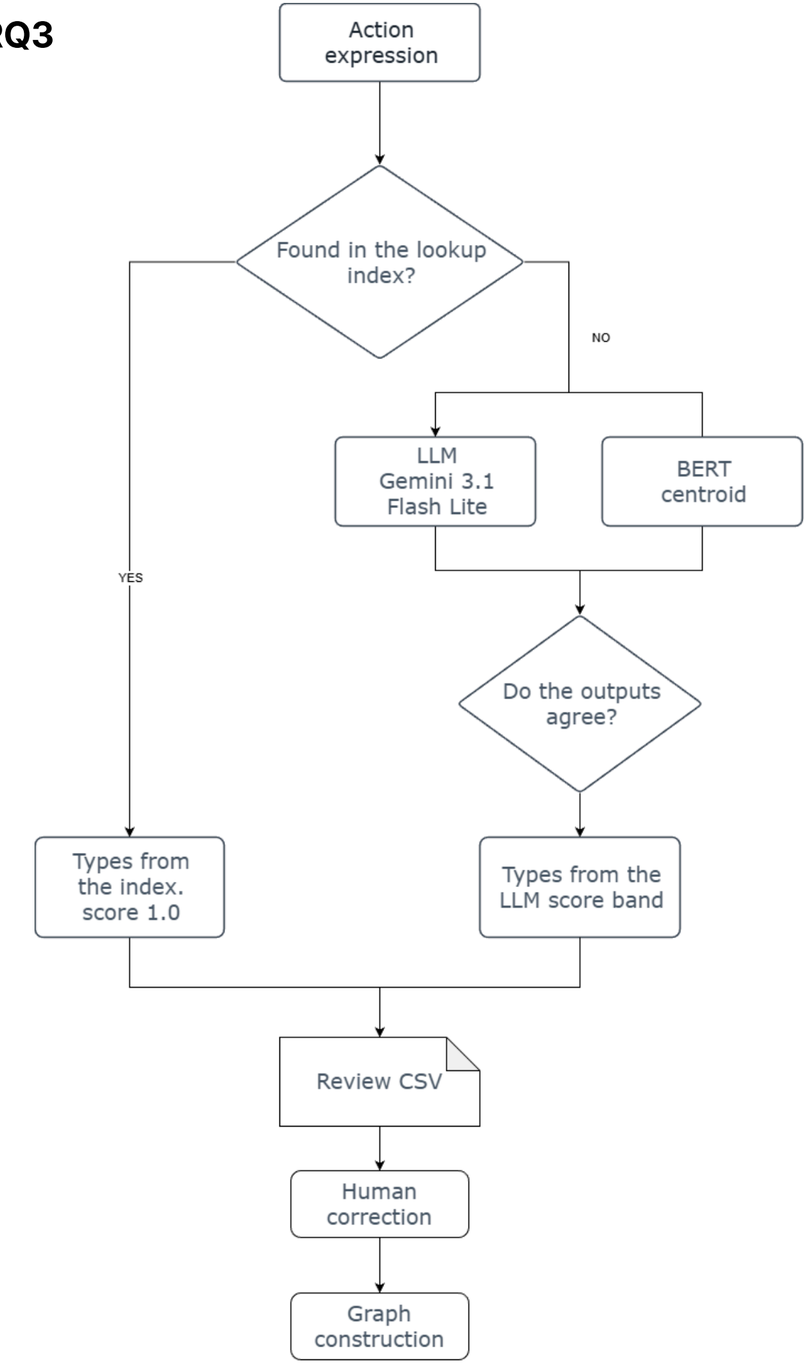
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RQ3

## ACT pipeline architecture



**RQ3**




Classification cascade with score and band

| State | Types assigned | Score | Band |
|---|---|---|---|
| Expression found in the lookup index | Lookup index | 1 | exact |
| LLM and BERT agree (the first type from BERT is among the LLM types) | LLM | 0.75 | strong |
| LLM and BERT both return types but disagree | LLM | 0.55 | moderate |
| Only the LLM returns types | LLM | 0.5 | weak |
| Only BERT returns types | BERT | 0.3 | none |
| Neither returns a type | — | 0.15 | none |

# Comparison of different action types over time

Temporal trend of # of events classified under the category **[Diplomacy & Relations]** and **[Military, Violence & Suppression]** in ***Pictorial of World History.***




**What this tells…**

- **A pattern of actions as recorded in a historical textbook made in Japan**
- **Raises further hypotheses: is there a correlation? is this a peculiarity of the source? is it due to availability of materials?**

# Comparison of actions taken by the same agent across sources

Action types of the events with Japan or the United States as agent, in two different sources

